Joseph Farkas

(770) 990-5495 | josephfarkas@gatech.edu | linkedin.com/in/farkasjoseph | github.com/farkasjoseph | US Citizen

Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Science, Devices and Intelligence *Expected May 2027*
Minors in Robotics and German. Concert Orchestra Member. *GPA: 4.00*

- Relevant Coursework: Data Structures and Algorithms, Objects and Design, Robotics and Perception

Professional Experience

UX Assistant Aug 2025 – Present
Georgia Tech PACE High Performance Computing Center *Atlanta, GA*

- Supporting roughly 5,000 researchers with incident triage across high-performance computing systems
- Handling 20+ tickets weekly, resolving issues independently using Linux tools and debugging skills

Software Intern Aug 2024 – May 2025
Viasat Inc. *Duluth, GA*

- Developed C/C++ firmware and Typescript frontend for prototype three-axis satellite antennas
- Built algorithms for analyzing signal quality and atmospheric modeling; integrated CI pipelines for validation

Autonomous Systems Lead, Software Mentor Oct 2021 – Present
North Gwinnett Robotics *Suwanee, GA*

- Mentoring ˜15 students with limited experience to use C++ to process sensor input and control robots over CAN
- Integrated localization using computer vision (Apriltag fiducials) to power fully autonomous scoring
- Implemented on-the-fly pathfinding and obstacle avoidance using A* with kinematic heuristics and telemetry
- World Comp. Autonomous Award & Division Champions. **Rated #1 globally for autonomous performance**

Activites

Guidance, Navigation and Control Team Aug 2025 – Present
Propulsive Landers @ Georgia Tech *Atlanta, GA*

- Developing RL model to "seed" model predictive control (MPC) during flight
- Applying PID+LQR and nonlinear MPC to optimize rocket's fuel usage in trajectory for powered descent

RoboCup: Autonomous Software Team Aug 2025 – Present
RoboJackets @ Georgia Tech *Atlanta, GA*

- Building multi-agent pathfinding, obstacle avoidance and task allocation in C++ for 6v6 robot soccer
- Strengthening ROS stack to interface with telemetry, vision, and low-level control systems

Projects, Presentations & Hackathons

soφ: Real-time Adaptive Learning | *Nexhacks (largest hackathon on the East Coast)* | **Devpost** Jan 2025

- Crafted app for instructor-aligned practice problems with real-time difficulty scaling using Gemini, Flask & React.
- Engineered hallucination-resistant pipeline with RAG + symbolic verification and compression for < 3s latency

Holonomic Auto-Alignment using CV Localization | *C++, Java, ArUco* Aug 2023 – May 2025

- Applied second-order kinematics and jerk limiting to replicate S-curve profile for discretized holonomic drive
- Fused CV estimates from fiducials with filtered odometry using a state-space model for real-time robot location
- Allowed competition robot to align to any scoring position with < 1cm accuracy regardless of starting state

Foundations of Autonomous Programming | *Georgia FIRST Robotics Symposium* Sept 2024

- Presented at statewide symposium on methods for teaching autonomous programming to high school students

Natural ASL Communication | *Gwinnett Science, Engineering & Innovation Fair* Feb 2024

- Interpreted 40 ASL phrases and 5 moods using Google Mediapipe and CNNs for natural English translation

Technical Skills

Computer Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, Dart, MATLAB, SQL
Frameworks & Libraries: TensorFlow, YOLO, Jupyter, ROS, Matplotlib, OpenCV, ArUco, React
Developer Tools: Git, Linux, Docker, Google Firebase, Android Studio, AWS, Jira
Hardware Experience: Fusion 360, SolidWorks, Autodesk Inventor, 3D Printing, CAN Network, CNC Operation